Time series: Smoothing

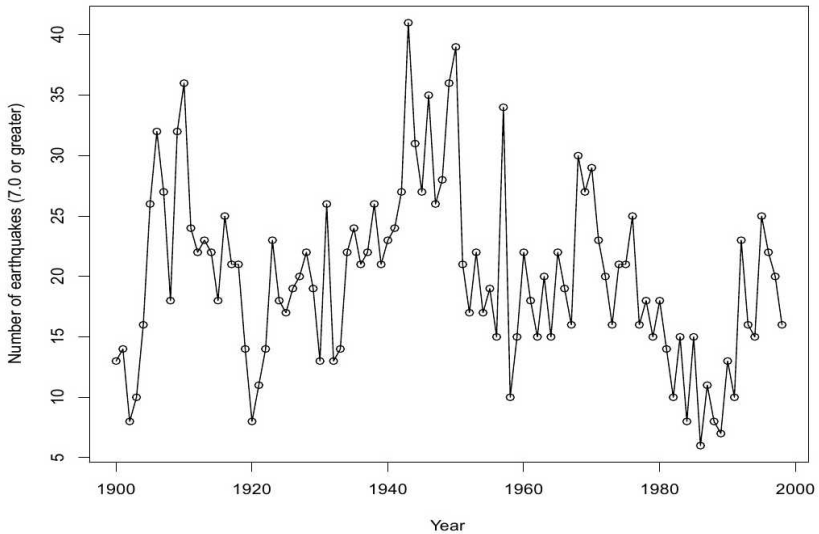
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The homework is provided on the final slides

Motivation for smoothing



The typical model of a time series is "signal + noise".

- **Goal:** To suppress the "noise". In other words, we want to estimate the underlying "signal".
- **Application:** Essential for decomposition of a given time series onto the sum of two time series:
  - "signal", a low-frequency component,
  - "noise", a high-frequency component.
- **Visual Clarity:** Makes patterns easier to see by eye.

Simple Moving Average (SMA)

The most common method is the **Moving Average**.
For a window size $m \in \mathbb{N}$, the smoothed value S_k is

$$S_k = \frac{1}{m} \sum_{i=0}^{m-1} x_{k-i}.$$

$$S_1, S_2, \ldots, S_{k-m}, S_{k-m+1}, \ldots, S_{k-1}, \underbrace{S_k}_{||}, S_{k+1}, \ldots, S_n$$

$$x_1, \; x_2, \ldots, x_{k-m}, \overbrace{x_{k-m+1}, \ldots, \; x_{k-1}, \; x_k} \; , x_{k+1}, \ldots, x_n$$

- **Small** m: Follows the original data closely (retains more noise).
- **Large** m: Produces a smoother line but may lag behind rapid changes.
- Values of S_k for $k = 1, \ldots, m-1$ should be defined as NA or calculated using an alternative formula.

Numerical Exercise: SMA Calculation

Task: Compute the Simple Moving Average (SMA) with $m = 3$ for the following time series x_k:

$$\{3, 7, 2, 5, 4, 8, 6\}.$$

Calculations: we have $S_k = \frac{1}{3}(x_{k-2} + x_{k-1} + x_k)$.

- $S_3 = \frac{3+7+2}{3} = \frac{12}{3} = 4.00$
- $S_4 = \frac{7+2+5}{3} = \frac{14}{3} \approx 4.67$
- $S_5 = \frac{2+5+4}{3} = \frac{11}{3} \approx 3.67$

k	1	2	3	4	5	6	7
x_k	3	7	2	5	4	8	6
S_k	–	–	**4.00**	**4.67**	**3.67**	**5.67**	**6.00**

Note: The first $m - 1$ values (for $k = 1, 2$) cannot be calculated with a backward-looking window of size 3.

R: SMA for simulated time series

We simulate a time series with a logistic trend and white noise.

```
# 1. Simulate data
t <- 1:100
trend <- 10 / (1 + exp(-0.15 * (t - 50))) # Logistic curve
noise <- rnorm(100, sd = 0.5) # Gaussian white noise
x <- trend + noise

# 2. Apply Simple Moving Average (n=10)
library(TTR)
ySMA <- SMA(x, n = 10)

# 3. Plot the results
plot(x, type = "l", col = "gray", main = "SMA on Logistic Trend")
lines(trend, col = "blue", lty = 2, lwd = 1) # True trend
lines(ySMA, col = "red", lwd = 2) # Smoothed signal
legend("topleft", legend=c("Observed", "True", "SMA"),
col=c("gray", "blue", "red"), lty=c(1,2,1))
```

Centered Moving Average

Another method is the **Centered Moving Average**. For a window size $2m+1$, the smoothed value C_k is

$$C_k = \frac{1}{2m+1} \sum_{i=-m}^{m} x_{k+i}.$$

- The smoothness of C_k increases as m increases.
- **Centered MA** uses values from both the past and the future to avoid time-lag.
- Values of C_k for $k = 1, \ldots, m, n-m+1, \ldots, n$ should be defined as NA or evaluated by another expression.

Numerical Exercise: Centered Moving Average

Task: Compute the Centered Moving Average (CMA) with window size 3 for the following time series x_k:

$$\{3, 7, 2, 5, 4, 8, 6\}.$$

Calculations: For a centered window of length 3, we have $C_k = \frac{1}{3}(x_{k-1} + x_k + x_{k+1})$.

- $C_2 = \frac{3+7+2}{3} = \frac{12}{3} = 4.00$
- $C_3 = \frac{7+2+5}{3} = \frac{14}{3} \approx 4.67$
- $C_4 = \frac{2+5+4}{3} = \frac{11}{3} \approx 3.67$

k	1	2	3	4	5	6	7
x_k	3	7	2	5	4	8	6
C_k	–	**4.00**	**4.67**	**3.67**	**5.67**	**6.00**	–

Note: We lose the first and last observations ($k = 1$ and $k = 7$) because the window cannot be centered at those points.

Weighted Moving Average

A general method is the **Weighted Moving Average**. For a window size $2m+1$ with coefficients $w_{-m}, \ldots, w_m \in \mathbb{R}_+$, the smoothed value W_k is

$$W_k = \frac{1}{\sum_{j=-m}^{m} w_j} \sum_{i=-m}^{m} w_i x_{k+i}.$$

Simple Exponential Smoothing (SES)

SES is used for series with no clear trend or seasonality. It gives more weight to recent observations as follows:

$$E_k = \alpha x_k + (1 - \alpha) E_{k-1},$$

where α is the **smoothing parameter**, $\alpha \in (0, 1)$.

- If α is close to 1, the model reacts quickly to changes.
- If α is close to 0, the model is very smooth and reacts slowly.
- Can be extended to **Holt-Winters** for data with trends and seasonality, see Slide 11.

Smoothing in R

```
library(fma) # for 'milk' time series
data("milk", package = "fma")
milkTS <- milk

m <- 15 # Simple Moving Average
library(TTR)
smoothMA <- SMA(milkTS, m)
plot(milkTS, type="l")
lines(smoothMA, col="red")
mtext(paste0("SMA m=", m), 3, 0)

alpha <- 0.2 # Exponential Smoothing
smoothES <- HoltWinters(milkTS, alpha, beta=FALSE,
gamma=FALSE)
plot(smoothES)
mtext(paste0("SES alpha=", alpha), 3, 0)
```

Holt-Winters for Seasonal Time Series

The Holt-Winters method extends exponential smoothing to capture both **trend** and **seasonality** simultaneously.

- **Additive Model:** Used when seasonal variations are roughly constant throughout the series.
- **Multiplicative Model:** Used when seasonal variations change proportionally to the level of the series.

The model consists of three smoothing equations for level (L_t), trend (b_t), and seasonal component (s_t):

$$
\begin{aligned}
L_t &= \alpha(x_t - s_{t-m}) + (1-\alpha)(L_{t-1} + b_{t-1}) \\
b_t &= \beta(L_t - L_{t-1}) + (1-\beta)b_{t-1} \\
s_t &= \gamma(x_t - L_t) + (1-\gamma)s_{t-m}
\end{aligned}
$$

where m is the seasonal period (e.g., 12 for monthly data). Think of L_t as the current 'center' of the data.

Holt-Winters in R

In R, the `HoltWinters()` function automatically estimates the smoothing parameters α, β, γ by minimizing the squared prediction errors.

```
# Fit Holt-Winters additive model
HWmodel <- HoltWinters(milkTS, seasonal = "additive")

# Plot the original data and the fitted values
plot(HWmodel)

# Forecast for the next 12 months
library(forecast)
plot(forecast(HWmodel, h=12))
```

Use of smoothing in statistical practice

- SMA and SES are typically used for smoothing in real-time (online) monitoring.
- CMA and WMA are popular for smoothing time series without periodicities (in offline mode).
- The HW method is very popular in econometrics for monthly time series.

Monday's challenging homework

- Use AI to learn how to record video using Zoom.
- Use AI to learn how to create a YouTube channel and upload videos.
- Create a secondary Google account and a corresponding YouTube channel without using your surname and student number.
- Record and upload a video demonstrating the use of the AI prompt that requires writing an R script for forecasting U.S. milk production 24 months ahead using the additive Holt-Winters model and executing the script in RStudio.
- Write the link to your video in the 'Discussions' tab on Learning Central.

Homework

- Use AI tools to find the R command for a "centered moving average".
- Use AI tools to write the R scripts for smoothing of time series without external libraries.
- Use AI tools to write the R scripts for solving exercises on slides 4 and 7.
- Apply SMA with $k = 5$ and $k = 20$ to the datasets from Lecture 1.
- Compare which k provides a better "seasonal adjustment" for monthly time series.
- **Math Task:** Prove that if $\alpha = 1$ in SES, the smoothed series is identical to the original series.